

Project Assignment

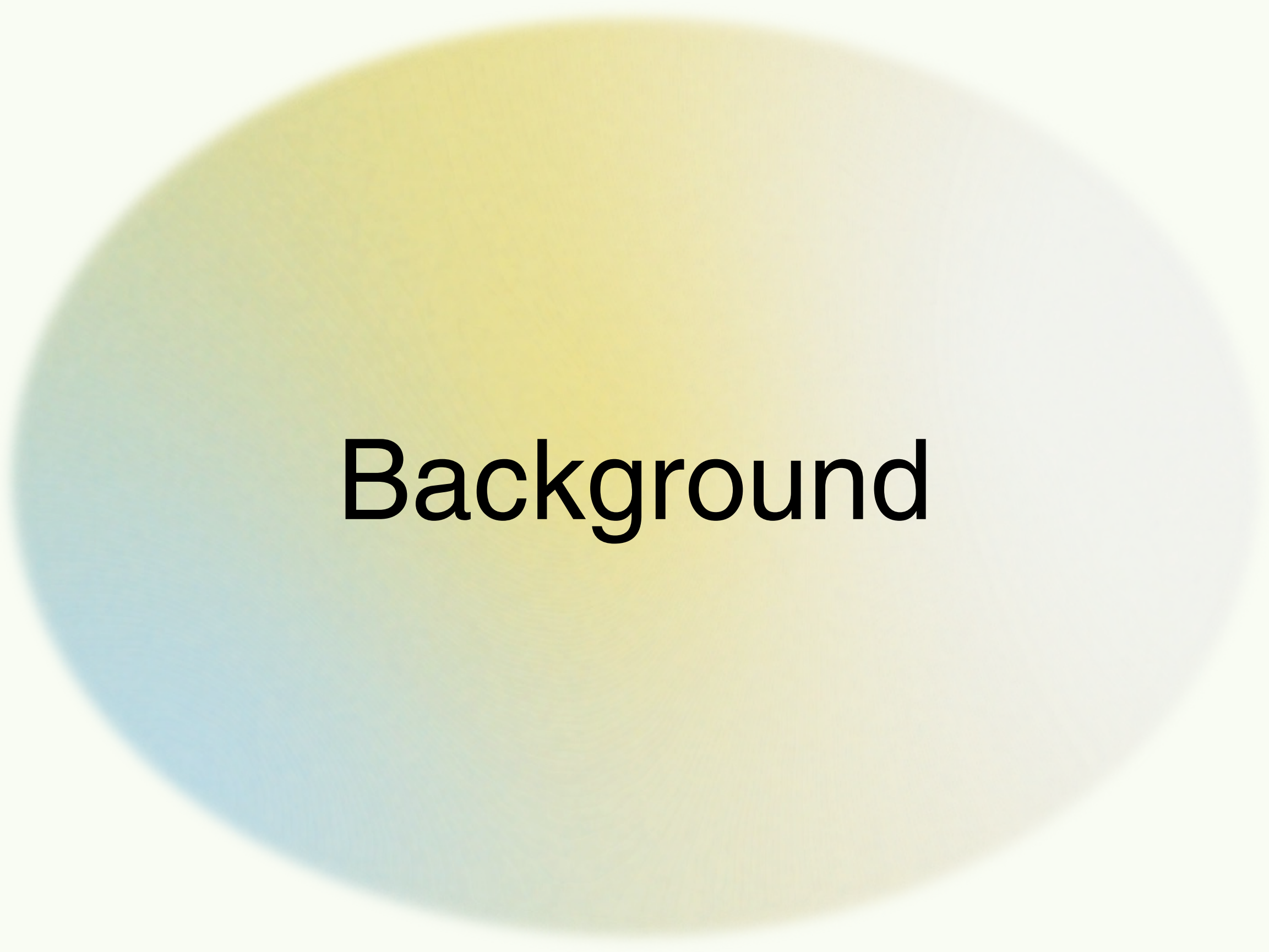
Description & deliverables



Agenda

Project assignment

- Background
- Datasets
- Task
- Evaluation & deliverables
- Group registration
- Polices



Background

CLAY

Recent paper (accepted at CVPR
2026, so still to be published!)

CLAY: Conditional Visual Similarity Modulation in Vision-Language Embedding Space

Sohwi Lim Lee Hyoseok Jungjoon Park Tae-Hyun Oh

KAIST



Figure 1. Our proposed concept-based conditional image retrieval method retrieves images focusing on the semantic aspects specified by the text condition. Given a query image, our method adaptively computes conditioned similarity by modulating the similarity space to align with various conditions, e.g., species, location, action, category, and color.

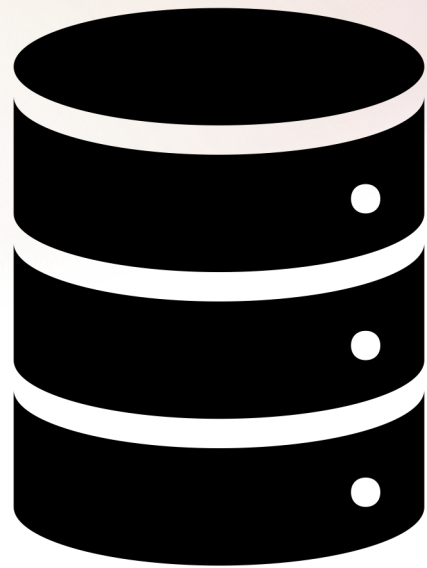
Abstract

Human perception of visual similarity is inherently adaptive and subjective, depending on the users' interests and focus.

works. Our code and datasets will be publicly released upon acceptance.

1. Introduction

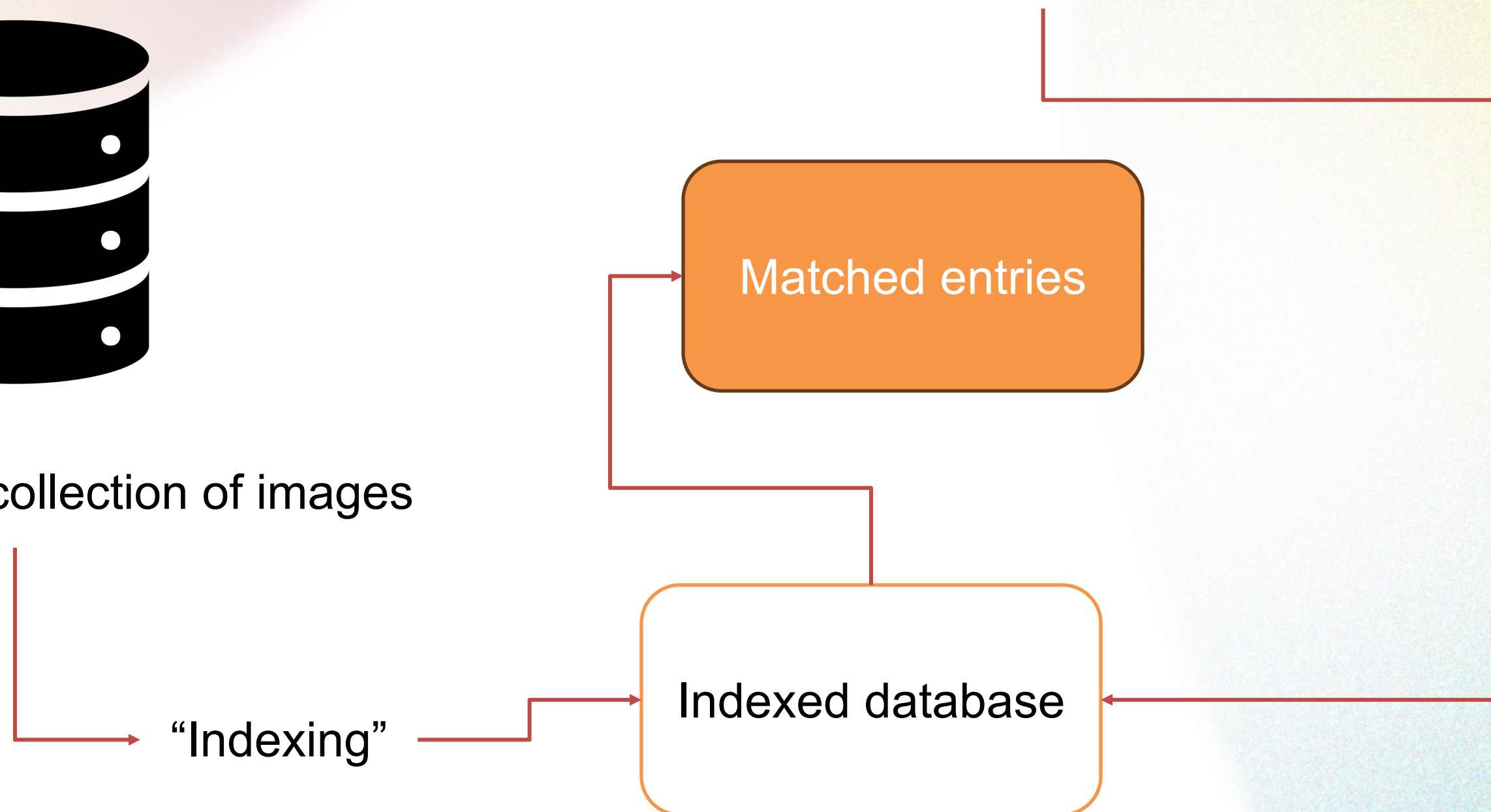
Retrieval



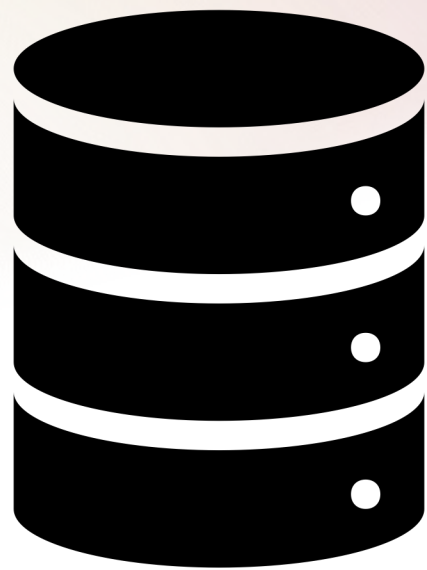
Database, *i.e.*, a collection of images



User query



Retrieval



Database, *i.e.*, a collection of images

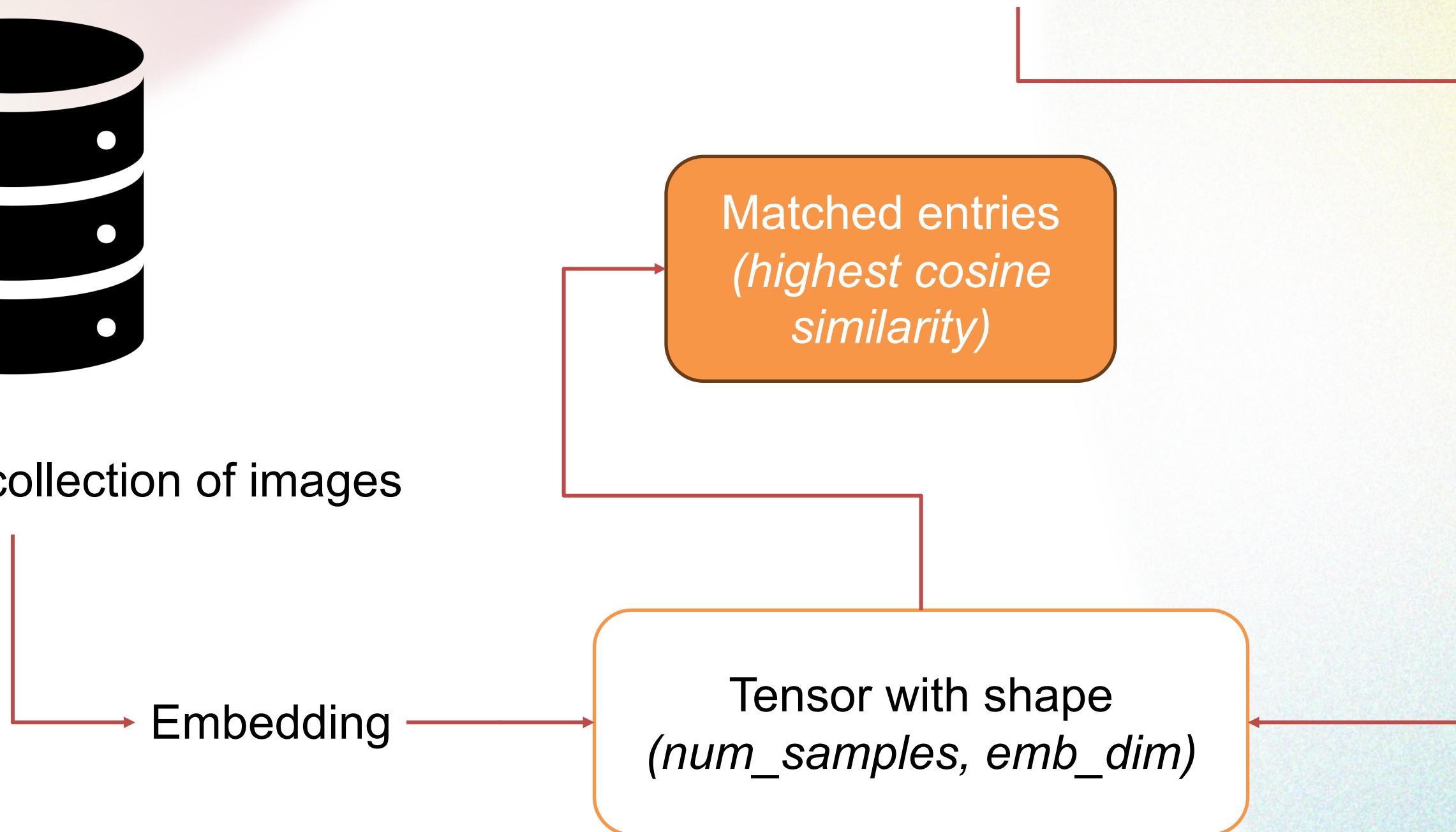


User query

Matched entries
(*highest cosine
similarity*)

Embedding

Tensor with shape
(*num_samples, emb_dim*)

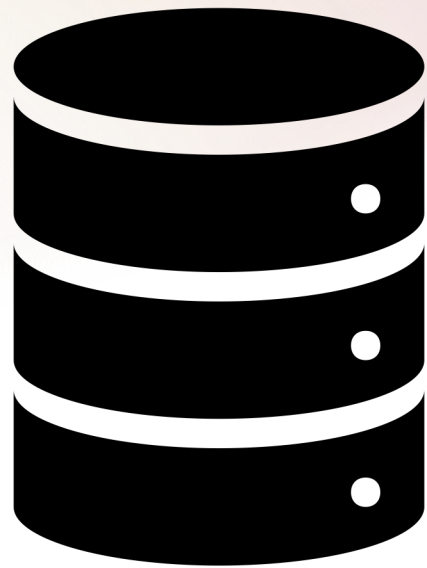


Retrieval



User query

If you change the embedding model, you need to re-embed the whole database, which can be very expensive!

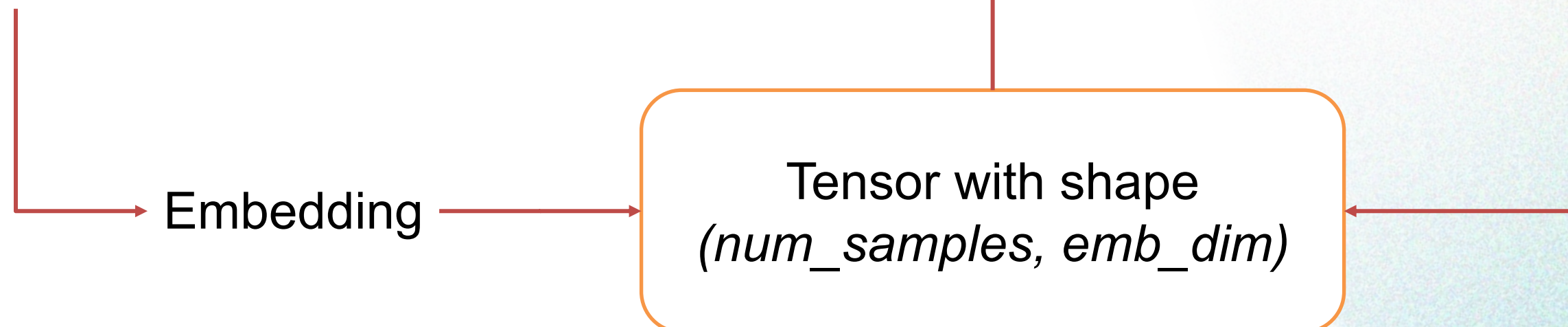


Database, *i.e.*, a collection of images

Embedding

Matched entries
(*highest cosine similarity*)

Tensor with shape
(*num_samples, emb_dim*)

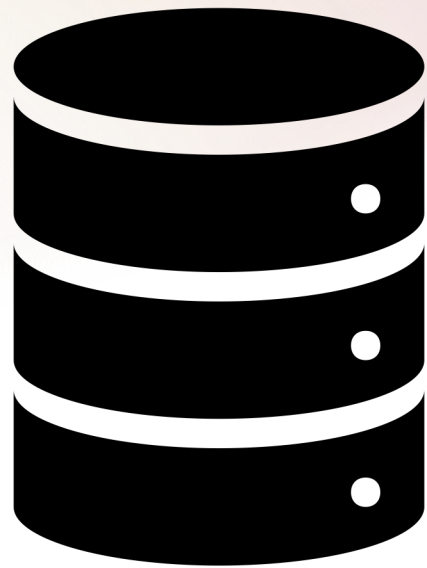


Retrieval



User query

Idea: instead of changing the entire model, we tweak the query encoder, keeping it compatible with the other encoder, so that we can keep the original database!



Database, *i.e.*, a collection of images

→ Embedding

Matched entries
(*highest cosine similarity*)

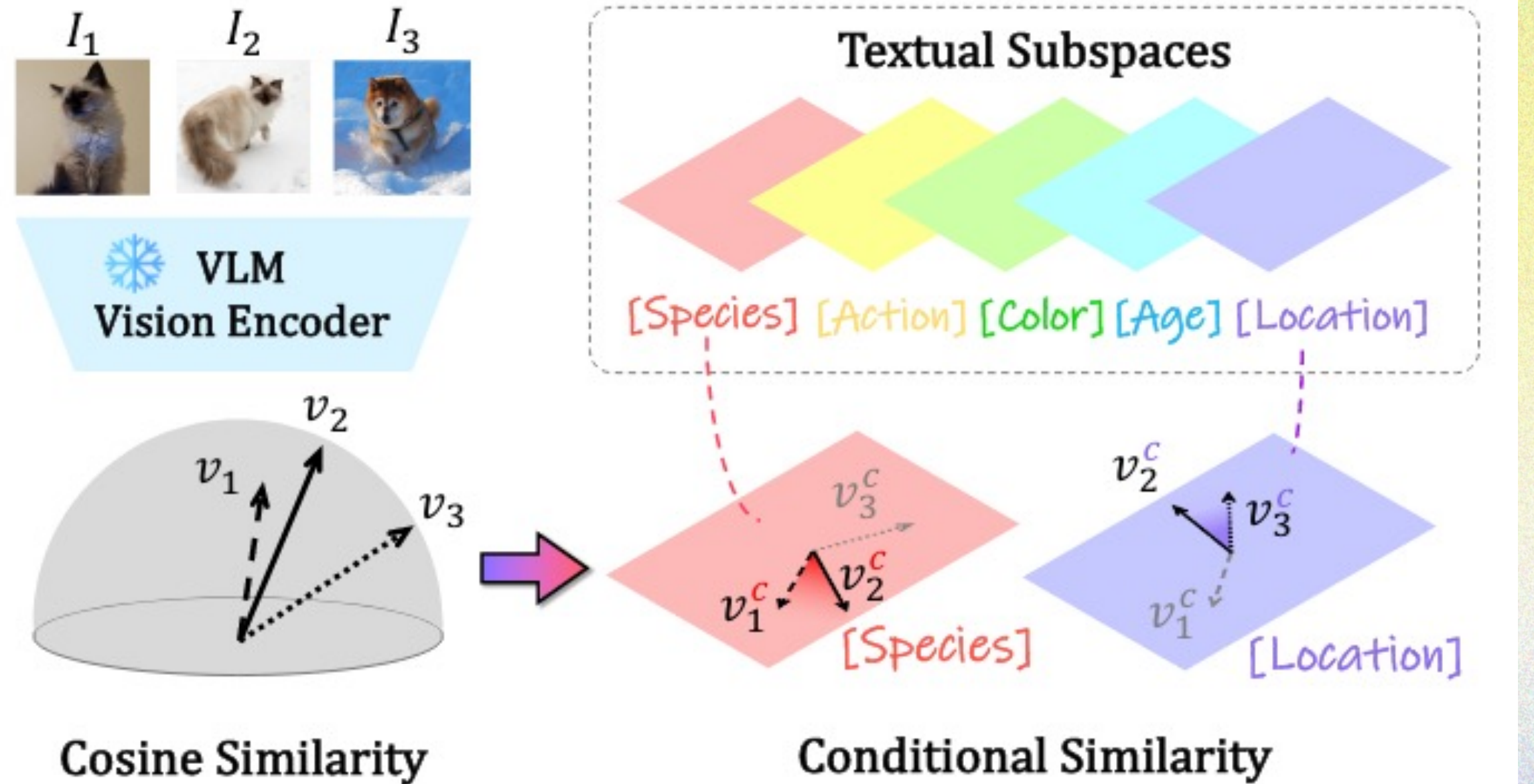
Tensor with shape
(*num_samples, emb_dim*)

→

←

CLAY

CLAY conditions the similarity between images by modulating the original similarity



CLAY

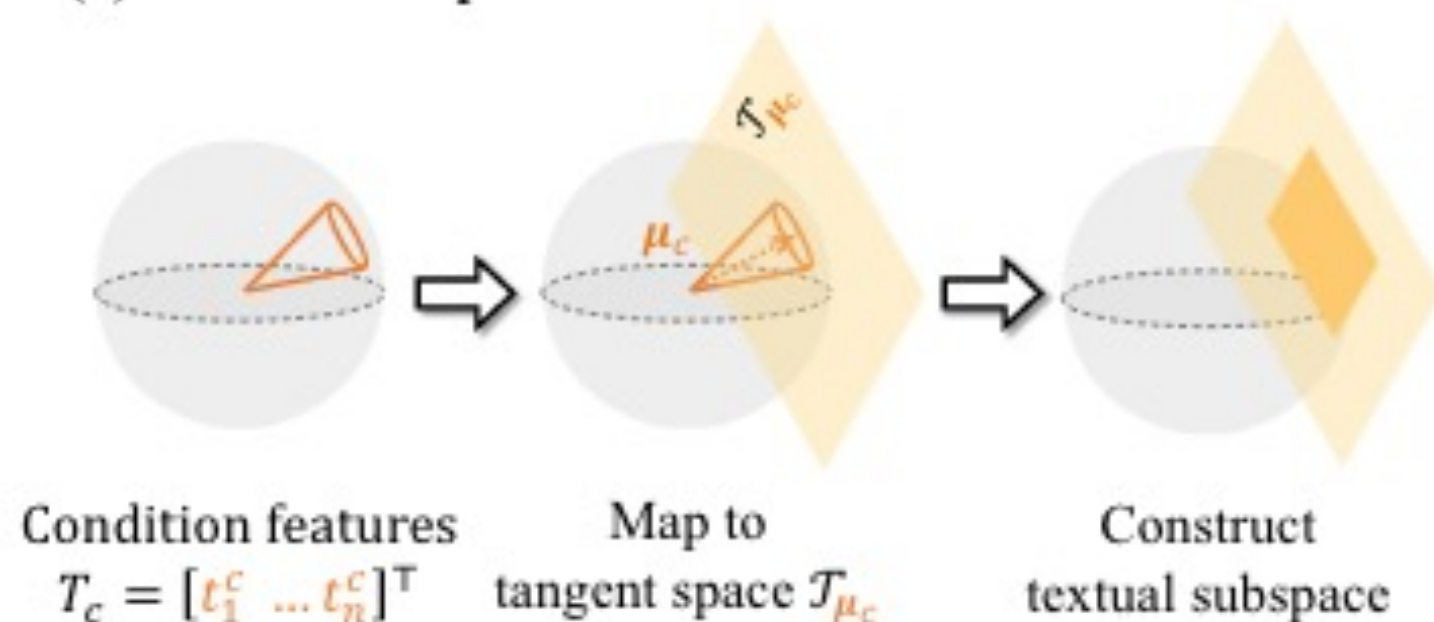
To keep the database intact, we need an *asymmetric* modulation:

$$\text{csim}_{\text{sym}}(I_q, I_d \mid c) = d(m(I_q, c), m(I_d, c)), \quad (1)$$

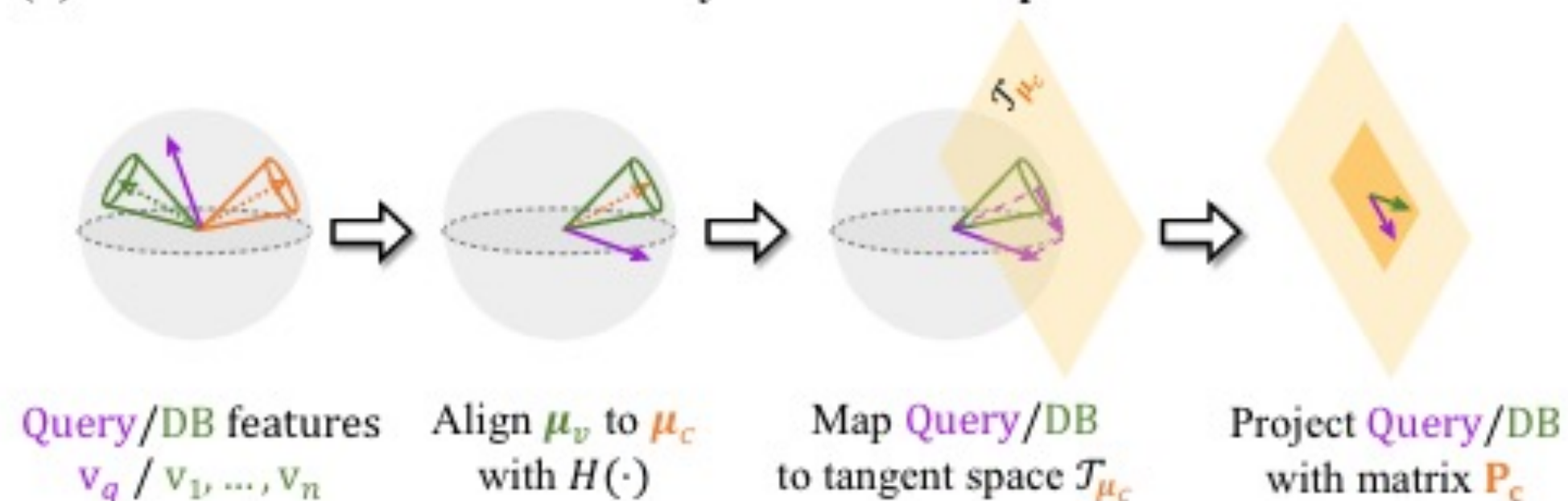
$$\text{csim}_{\text{asym}}(I_q, I_d \mid c) = d(m(I_q, c), I_d), \quad (2)$$

CLAY

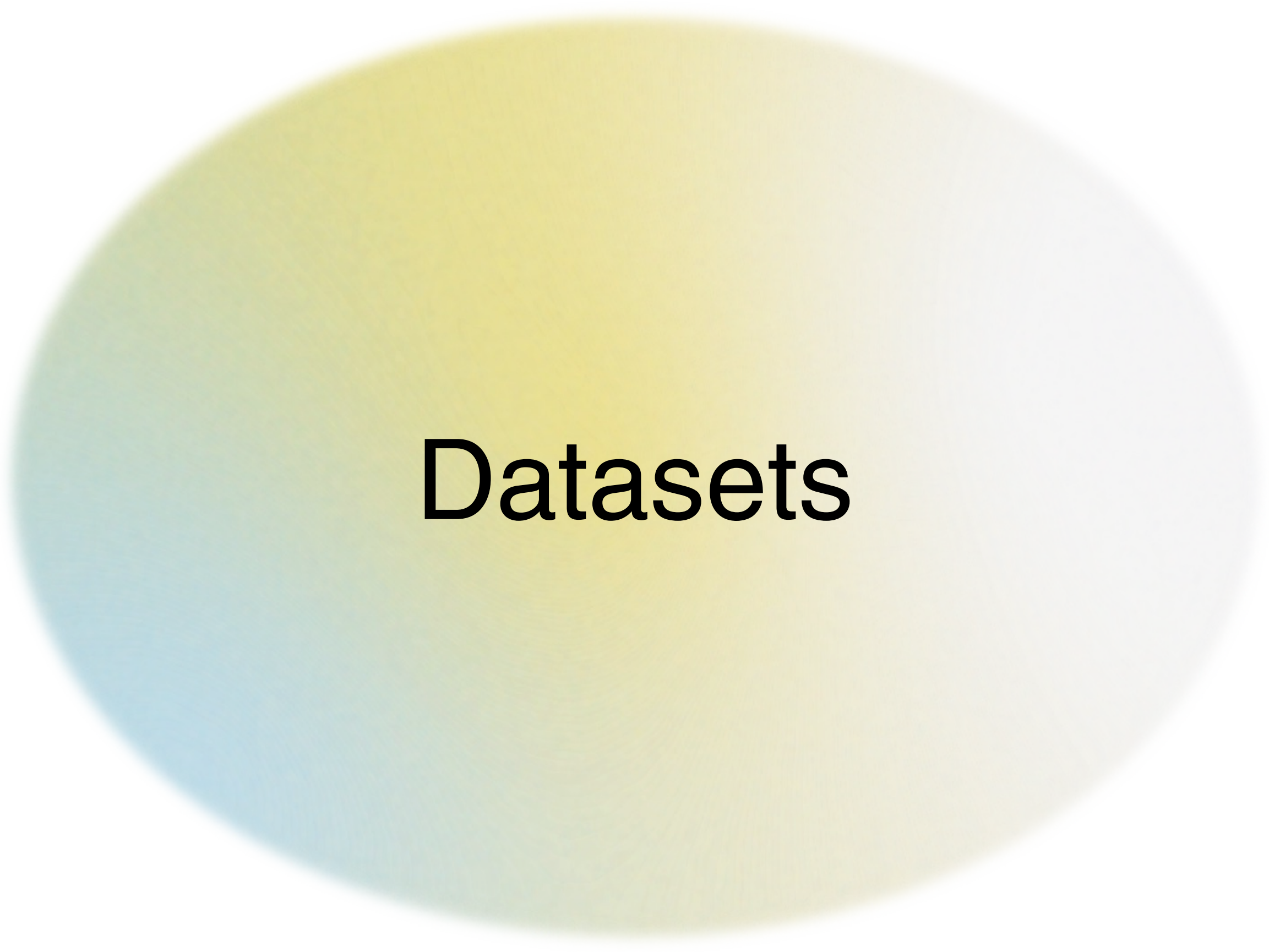
(a) Textual subspace construction



(b) Inference: Conditional Similarity on textual subspace



: Textual / Visual features
 : Mean of features
 : Tangent space of μ_c
 : Textual subspace
 : Query feature



Datasets

Datasets

You will work on CELEB-A

40 binary attributes for each image, allowing you to experiment easily

Eyeglasses



Wearing Hat



Bangs



Wavy Hair



Pointy Nose



Mustache

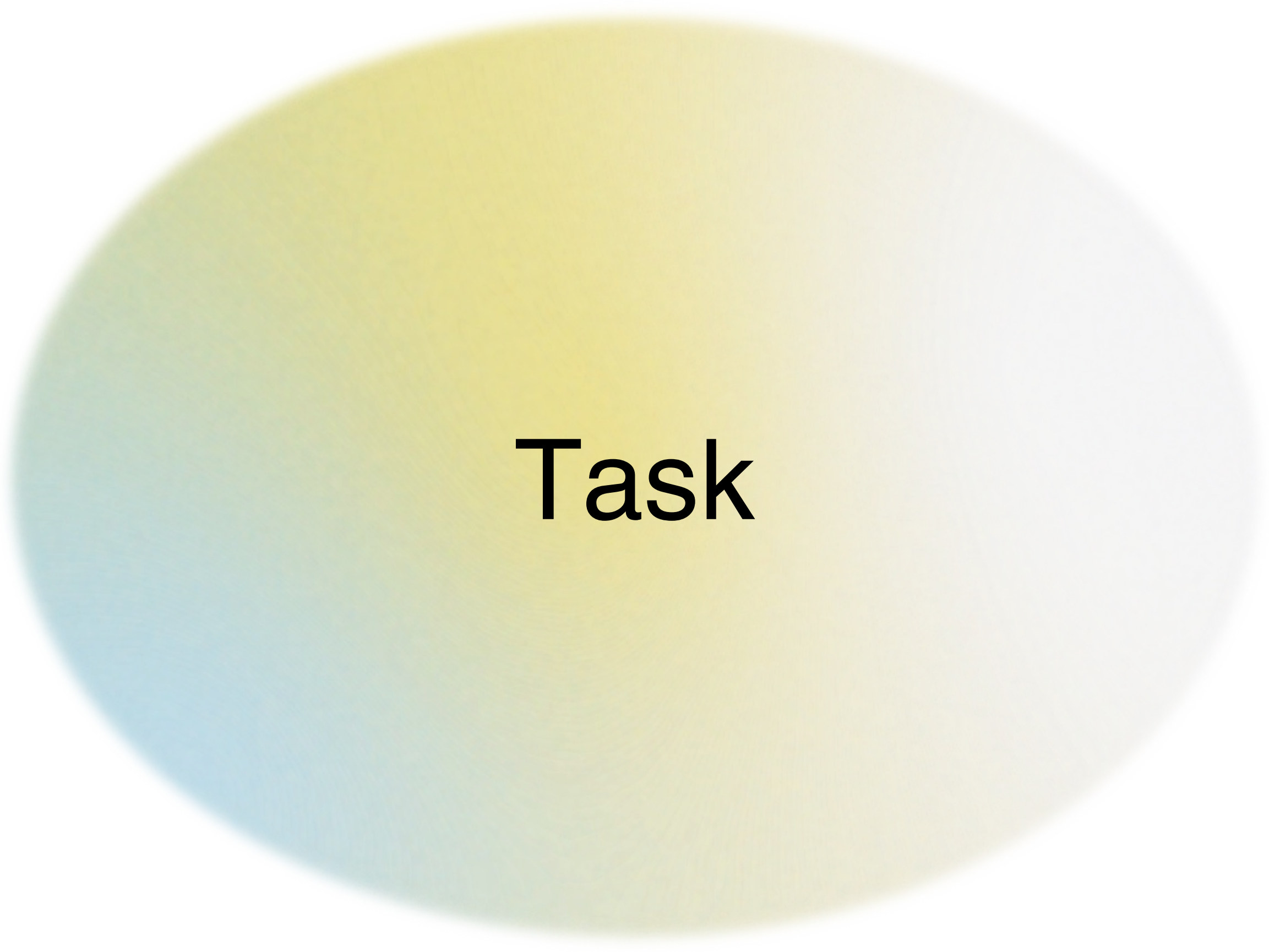


Oval Face



Smiling





Task

Task description

CLAY has a rigid pre-SVD embedding
stacking as a fusion mechanism.

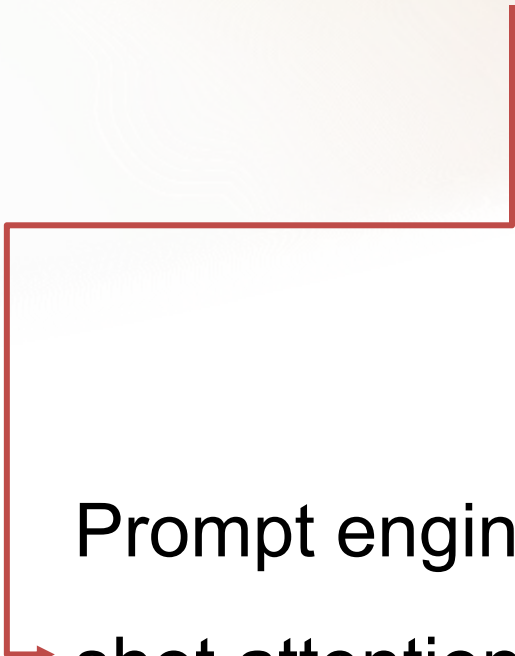
You should develop a new
methodological approach.

No restriction: you can
do *whatever* you want!

Task description

Training-free approaches

Training-based approaches



Prompt engineering, zero-shot attention scaling,
latent space arithmetic



Lightweight adapters, cross-attention networks, interpretability-based sparse autoencoders (SAEs)

Task description

Main challenges

Expressive multimodal conditioning

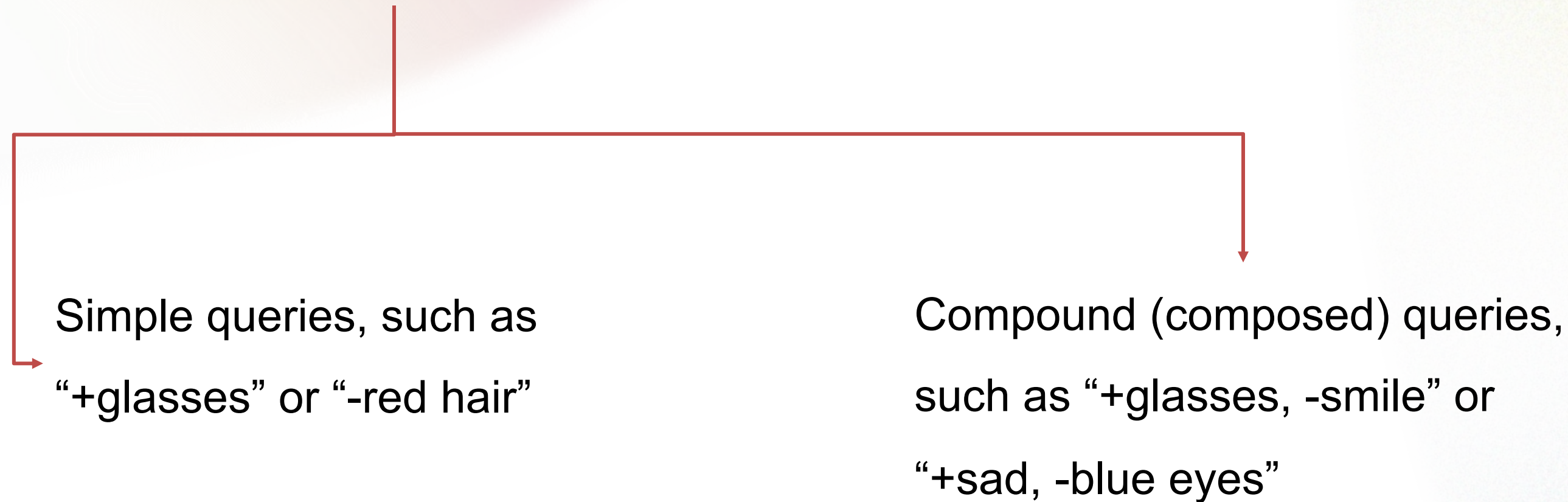
- Accept one or multiple conditions as input, together with the given image
- Accept both *positive* and *negative* conditions

Overcoming the SVD bottleneck

- CLAY uses SVD, but it limits the expressiveness of the model
- Explore diverse mechanisms!

Evaluation protocol

We will provide you with queries



You will find them in the document. We will update it shortly!

Evaluation protocol

To make things simple and fair, we have defined a set of queries. For each query, we checked that there existed images that would satisfy it.

We will provide you with a JSON file that contains all the queries, the images that you should use as “source” for each query, and the candidate targets (*i.e.*, the ground truth) for each source.

Evaluation protocol

To make things simple and fair, we have defined a set of queries. For each query, we checked that there existed images that would satisfy it.

We will provide you with a JSON file that contains all the queries, the images that you should use as “source” for each query, and the candidate targets (*i.e.*, the ground truth) for each source.

We will also provide the evaluation code, so that it's easy to compare results across groups – both for you and for us!

Metrics

The code we will provide evaluates Recall@K and Precision@K:

$$Recall@K = \begin{cases} 1, & \text{if } |\mathcal{R}_K \cap \mathcal{G}| > 0 \\ 0, & \text{otherwise} \end{cases}$$

$$Precision@K = \frac{|\mathcal{R}_K \cap \mathcal{G}|}{K}$$

Roadmap

We suggest you follow this roadmap:

1. Data exploration and preprocessing
2. Offline feature extraction
3. Vanilla zero-shot baseline
4. Method development



**Evaluation &
deliverables**

What we expect

A *single* Jupyter Notebook, hosted on Google Colab

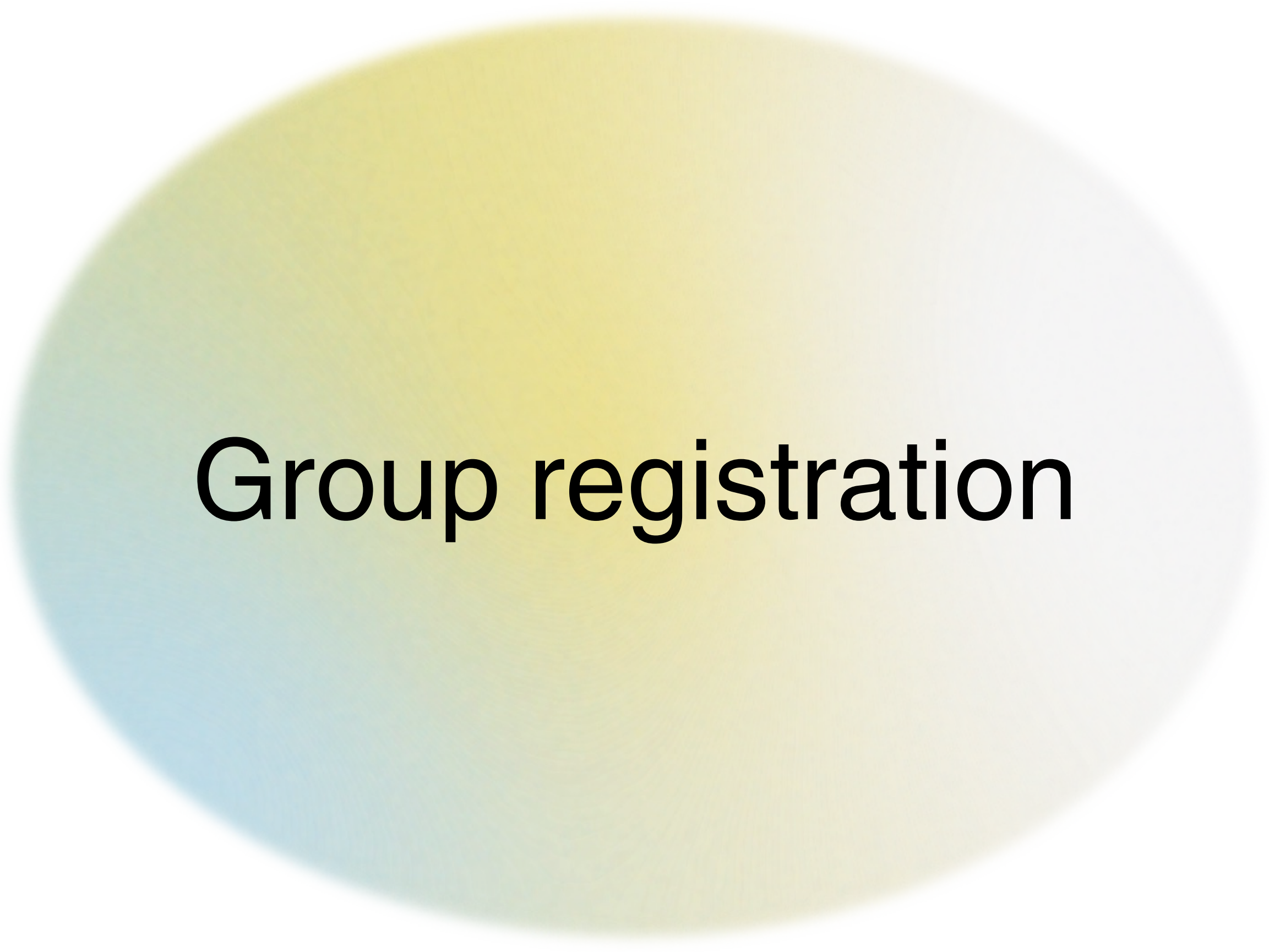
It should be self-contained. It must have:

- Your code, so that we can run it
 - Clean and readable! Leave comments
- Leave output cells, so that we can check it more easily
- A *comprehensive* report, in the Markdown cells between code cells
 - Methodological description: a detailed description of the solution you are proposing
 - Experimental setup: describe your training / evaluation strategy, and motivate all your choices!
 - Results and discussion: make sure we understand why you got your results

Evaluation axes

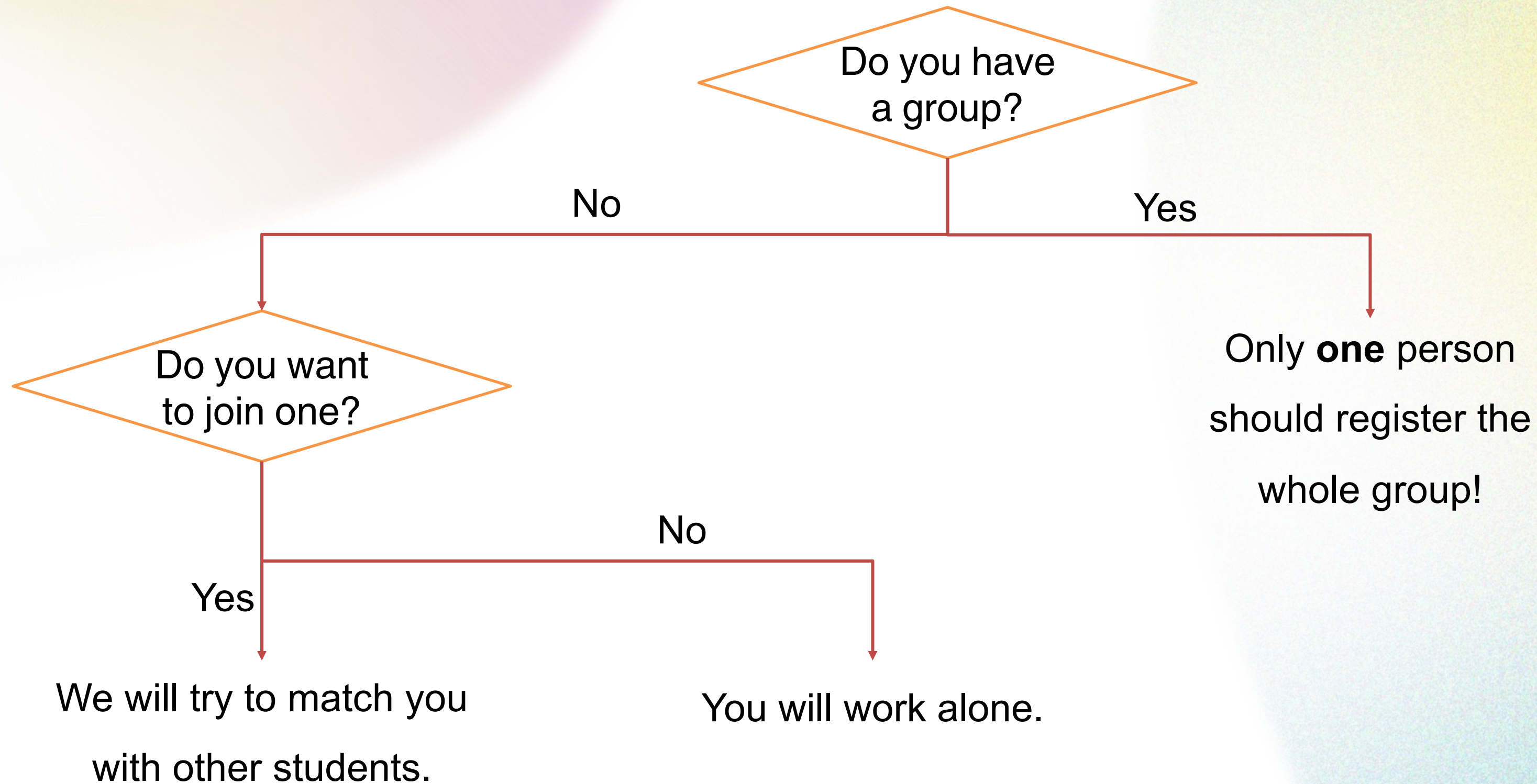
We will evaluate your work based on:

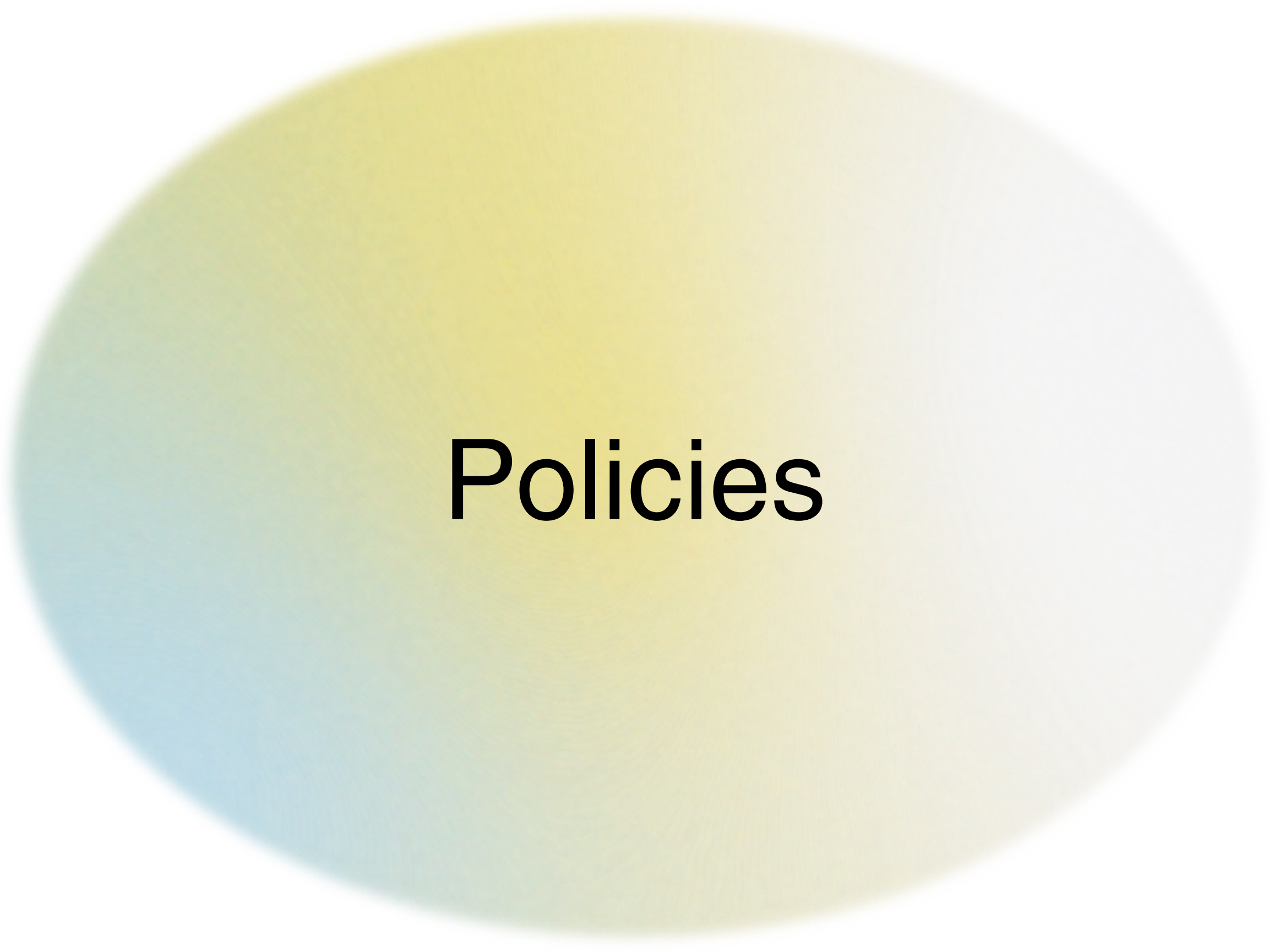
1. Originality and creativity
2. Methodological thoroughness
 - Don't just say "we did this and it worked"
3. Report clarity
4. Empirical performance
 - We'd like to see some decent performance, but we're not interested in you becoming SOTA. Just show you can improve upon a baseline!
5. Code quality
 - Don't submit bad code



Group registration

Group registration





Policies

Policies

- *Do not* copy-paste code from other repositories
 - We want you to understand what you are doing and how PyTorch works. Don't use code as a black-box!
- You can *borrow* code from the Internet, such as “standard” functions, but cite the source!
- You *are* encouraged to discuss your approach with other groups
- You *should not* share your code with other groups!

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Description & deliverables